

1 Gain law

(bars denote normalized quantities: lengths by R , $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}'(\bar{a}) := \frac{d\bar{a}}{d\bar{w}} = (1 - \bar{a})^2, \quad a_w = a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R}$$

$$\bar{a}(\bar{w}) = 1 - \frac{1}{\bar{w} - \bar{w}_0}, \quad \frac{\partial \bar{a}}{\partial \bar{w}_0} = -\bar{a}'$$

$$b_{\text{state}}(\bar{w}) := \frac{(1 - \bar{a}_{\text{true}})^2}{\bar{a}'_{\text{true}}}, \quad \frac{\bar{a}'(\bar{a}_{\text{true}})}{\bar{a}'_{\text{true}}} - 1 = b_{\text{state}}(\bar{w}) - 1$$

2 Jacobians

$$\frac{\partial \bar{a}'}{\partial \bar{a}} = -2(1 - \bar{a}), \quad \frac{\partial}{\partial \delta a} [\bar{a}_w + (1 - \bar{a}_w)^2(\cdot) + \delta a] = 1$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\begin{aligned} \bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k], & \bar{w}_d[k + 1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\ \delta \bar{w}_1[k + 1] &= \delta \bar{w}_2[k], & \delta \bar{w}_2[k + 1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k + 1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \end{aligned}$$

$$\begin{aligned} F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i] \\ \varepsilon_w[k] &= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] + (1 - \lambda_c) n_w[k] \end{aligned}$$

$$\bar{a}_w[k + 1] = \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

(a)

$$\bar{a}_w[k + 1] = \bar{a}_w[k] + (1 - \bar{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

(b)

$$\bar{a}_w[k+1] = \bar{a}_w[k] + (1 - \bar{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] + \delta a[k]$$

$$\begin{aligned} \delta a[k] &= \left[\bar{a}'_{\text{true}}[k] - (1 - \bar{a}_w[k])^2 \right] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] \\ &= (1 - \bar{a}_w[k])^2 \left[\frac{1}{b_{\text{state}}(\bar{w}[k])} - 1 \right] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] \end{aligned}$$

$$\delta a[k+1] = \delta a[k]$$

4 Process model

$$(a) \quad \mathbf{x} = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w]^\top, \quad (b) \quad \mathbf{x} = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, \delta a]^\top \quad (d=2)$$

5 Estimator

$$\delta \hat{w}_1[k+1] = \delta \hat{w}_2[k], \quad \delta \hat{w}_2[k+1] = \delta \hat{w}_3[k], \quad \delta \hat{w}_3[k+1] = \lambda_c \delta \hat{w}_3[k]$$

$$\begin{aligned} (a) \quad \hat{a}_w[k+1] &= \hat{a}_w[k] + (1 - \hat{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right] \\ (b) \quad \hat{a}_w[k+1] &= \hat{a}_w[k] + (1 - \hat{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right] + \delta \hat{a}[k], \quad \delta \hat{a}[k+1] = \delta \hat{a}[k] \end{aligned}$$

6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$\begin{aligned} e_{\delta \bar{w}_1}[k+1] &= e_{\delta \bar{w}_2}[k], \quad e_{\delta \bar{w}_2}[k+1] = e_{\delta \bar{w}_3}[k] \\ e_{\delta \bar{w}_3}[k+1] &= \lambda_c e_{\delta \bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ e_{\bar{a}_w}[k+1] &= \left[1 + (1 - \hat{a}_w)^2 F_{dw} - 2(1 - \hat{a}_w)(\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) \right] e_{\bar{a}_w}[k] \\ &\quad + (1 - \lambda_c)(1 - \hat{a}_w)^2 e_{\delta \bar{w}_3}[k] + (1 - \hat{a}_w)^2 \varepsilon_w[k] \quad [+ e_{\delta a}[k] \text{ in (b)}] \\ e_{\delta a}[k+1] &= e_{\delta a}[k] \end{aligned}$$

$$(b) \quad F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1 - \lambda_c)(1 - \hat{a}_w)^2 & 1 + (1 - \hat{a}_w)^2 F_{dw} - 2(1 - \hat{a}_w)(\Delta \bar{w}_d + (1 - \lambda_c)\delta \hat{w}_3) & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(a) is the leading 4×4 block.

$$\mathbf{q}[k] = [0, 0, -\varepsilon_w, (1 - \hat{a}_w)^2 \varepsilon_w, 0]^\top$$

$$e_{\bar{a}}(\bar{w}) = -\bar{a}'(\bar{w}) e_{\bar{w}_0} \propto \frac{1}{(\bar{w} - \bar{w}_0)^2}$$

7 Measurement and innovation

$$\nabla_d \bar{w}_d[k] = \bar{w}_d[k] - \bar{w}_d[k - d]$$

$$y_1[k] = \delta \bar{w}_1[k] + n_w[k]$$

$$y_2[k] = \bar{a}_w[k - d] + n_a[k]$$

$$\bar{a}_w[k - d] = \bar{a}_w[k] - (1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k] \quad [-d \delta a[k] \text{ in (b)}]$$

$$\frac{\partial y_2}{\partial \bar{a}_w} = 1 + 2(1 - \hat{a}_w) \nabla_d \bar{w}_d, \quad \frac{\partial y_2}{\partial \delta a} = -d$$

$$(b) \quad H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2(1 - \hat{a}_w) \nabla_d \bar{w}_d & -d \end{bmatrix}, \quad K[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$e_{y_1}[k] = e_{\delta \bar{w}_1}[k] + n_w[k]$$

$$e_{y_2}[k] = [1 + 2(1 - \hat{a}_w) \nabla_d \bar{w}_d] e_{\bar{a}_w}[k] - d e_{\delta a}[k] + n_a[k]$$

8 Process and measurement noise

$$q_3 = -\bar{a}_w \bar{f}_T, \quad q_4 = -(1 - \hat{a}_w)^2 q_3, \quad \gamma(\bar{w}) = \gamma_N c(\bar{w}), \quad \sigma_{f_T}^2 = a_o^2 \frac{4k_B T \gamma(\bar{w})}{\Delta t}$$

$$Q_{33} = \frac{4k_B T a_o}{R} \bar{a}_w, \quad Q_{34} = Q_{43} = -(1 - \hat{a}_w)^2 Q_{33}, \quad Q_{44} = (1 - \hat{a}_w)^4 Q_{33}, \quad Q_{55} = 0$$

$$R = \text{diag}(R_1, R_2), \quad R_1 = \sigma_{n_w}^2, \quad R_2 = K_{\text{var}} I F_{\text{var}} (\bar{a}_w + \xi)^2 + d Q_{44}, \quad K_{\text{var}} = \frac{2a_{\text{cov}}}{2 - a_{\text{cov}}}$$

$$\sigma_{\delta \bar{w}_r}^2 = C_{dpmr} \frac{4k_B T a_o}{R} (\bar{a}_w + \xi), \quad \xi = \frac{C_n \sigma_{n_w}^2}{C_{dpmr} (4k_B T a_o / R)}$$

$$Q_{55}^* = \text{Var}(\delta a[k+1] - \delta a[k])$$

Seed and prior

$$\delta \hat{a}[0] = 0, \quad \hat{a}_w[0] = 1 - \frac{1}{\bar{w}[0] - \hat{w}_0}$$

$$\left. \frac{\partial \bar{a}}{\partial \delta a} \right|_{\bar{w}} = \frac{1 - \bar{a}}{1 + \delta a}, \quad \left. \frac{\partial \bar{a}}{\partial \bar{w}_0} \right|_{\bar{w}} = -\bar{a}'$$

$$P_{44}[0] = (\hat{a}'[0])^2 P_{\bar{w}_0} + \text{floor}_a^2, \quad P_{45}[0] = 0, \quad P_{55}[0] = P_{\delta a}$$